Tracer Adaptive LesionLocator Ensemble

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Abstract. We describe an interactive whole-body tumor segmentation system for FDG and PSMA PET/CT. A five-fold nnU-Net residual-encoder ensemble produces the initial mask, followed by tracer-specific calibration and a prompt-conditioned distance-transform model. We introduce conservative interaction changes: clicks are mapped to cropped model space using one crop-origin subtraction; donor components are validated independently to prevent a newly detected lesion from hiding a harmful bridge; PSMA foreground corrections are enabled up to a fixed 128-component burden; and explicitly annotated PSMA background voxels are removed only when removal cannot fragment a predicted component. Exact foreground-scribble voxels are also enforced without dilation when they do not merge accepted lesions. Paired causal replay on local interactive fixtures increased mean AUC-Dice from 0.8403 to 0.8443 and mean AUC-DMM/F1 from 0.9136 to 0.9139, without an observed per-case regression. The method is stateless, deterministic with respect to cumulative scribbles, and is released with complete upstream attribution.

Keywords: autoPET V · interactive segmentation · PET/CT · topology

1 Introduction

autoPET V evaluates lesion segmentation across FDG and PSMA PET/CT under iterative simulated and clinician-driven scribbles. High voxel overlap alone is insufficient: a useful method must also create missed lesion instances and remove false-positive instances without merging anatomically separate tumors. Our submission targets this topological failure mode while preserving strong initial segmentation from public AutoPET-III and AutoPET-IV components.

2 Methods

2.1 Data

Training data The pretrained components use the public data and preprocessing described by the upstream AutoPET-III LesionTracer and AutoPET-IV LesionLocator releases. Our changes introduce no additional training data or weight optimization.

Validation data Local validation used six FDG and four PSMA interactive fixtures, with cumulative challenge-format scribbles replayed for five steps. These fixtures were used for paired engineering comparisons and are not presented as an unbiased estimate of hidden-test rank.

2.2 Data preprocessing

CT and PET are resampled and normalized by the frozen nnU-Net plans. Predictions are returned to the CT reference grid, and the output copies its size, spacing, origin, and direction. Input-grid click coordinates are mapped into cropped model space by subtracting the crop origin once and then applying the crop-to-resampled scale. No metadata identifying the tracer is required.

2.3 Initial segmentation and tracer routing

Three frozen classifiers route each scan to FDG or PSMA. The initial mask is a five-fold AutoPET-III LesionTracer ResEncL ensemble without mirror test-time augmentation. FDG uses a probability threshold of 0.47 and burden-adaptive small-component filtering. PSMA uses threshold 0.50, five-voxel filtering, and the upstream frozen logistic component pruner at threshold 0.86.

2.4 Algorithm and topology-safe interaction

Later calls reconstruct the initial mask and run the fold-0 EDT LesionLocator with all cumulative scribbles. Each clicked foreground donor component is considered independently. It is accepted only if union with the current mask does not reduce the connected-component count, preventing lesion bridges. For PSMA, foreground correction requires at least six cumulative foreground points and at most 128 initial components. Each connected background stroke is also handled independently: only its explicitly marked voxels are erased, and the erase is accepted only when it does not increase component count. FDG background behavior remains unchanged from the upstream system. Finally, each connected foreground scribble stroke is treated as direct supervision: its exact marked voxels are added without dilation if doing so does not reduce component count. This last operation is independent of network probability calibration.

2.5 Data postprocessing

FDG predictions use a probability threshold of 0.47 and burden-adaptive connected-component filtering. PSMA uses threshold 0.50, five-voxel filtering, and a frozen logistic component pruner with false-positive threshold 0.86. All interactive unions and erasures are checked using 18-connected components.

2.6 Training and test parameters

All network weights are unchanged pretrained weights. Consequently, no new optimizer, schedule, augmentation, or training hardware is applicable to this derivative. Inference uses five initial-model folds, one interactive fold, disabled mirror TTA, binary connected components, and cumulative stateless scribble replay. Engineering and local validation used up to six NVIDIA H100 80 GB HBM3 GPUs. The submitted image targets an NVIDIA T4 with 16 GB VRAM.

2.7 GitHub repository

Code, tests, build recipes, notices, and checkpoint integrity hashes are released under a permissive license at github.com/marchmello1/Tracer-Adaptive-LesionLocator-Ensemble.

3 Results

Relative to the frozen v2 policy, exact foreground-stroke enforcement raised mean causal AUC-Dice from 0.8403 to 0.8443 and mean AUC-DMM/F1 from 0.9136 to 0.9139 over nine defined local cases. Dice improved in five cases and was unchanged in four; F1 improved in one and was unchanged in eight. The largest Dice change was PSMA3 (0.7937 to 0.8238), while PSMA1 F1 changed from 0.8871 to 0.8898. These local fixtures support selection of the policy but are not a substitute for hidden-test evaluation. The earlier frozen v2 container obtained preliminary-test Dice 0.854649, F1/DMM 0.834179, and mean position 3.5. The preliminary set contains only five cases and is reported as an implementation check, not as a final-test performance claim.

4 Discussion and conclusion

The method changes interaction logic rather than claiming newly trained model weights. Its main advantage is a simple invariant: one useful new lesion may not compensate for a separate harmful bridge in the same update. The 128-component ceiling is deliberately conservative and should be re-estimated on a larger independent validation set. Exact foreground-stroke enforcement reduces dependence on probability calibration, but the small validation cohort does not establish final-set performance.

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Disclosure of Interests The author declares no competing interests.

Table 1. Algorithm details.

Item	Configuration
Team	Nuvo AI
Algorithm	Tracer Adaptive LesionLocator Ensemble
Input and preprocessing	3-D whole-body PET/CT; nnU-Net normalization, resampling, and sliding-window inference
Initial segmentation	Five-fold residual-encoder U-Net ensemble with a three-model tracer vote
Interaction model	One prompt-conditioned EDT fold with cumulative stateless scribble replay
Postprocessing	Tracer calibration, component filtering, and topology-safe component fusion
Training and augmentation	Public pretrained models; no new training or augmentation
Test-time augmentation	None
Framework	nnU-Net v2 and PyTorch
Development hardware	Up to six NVIDIA H100 80 GB HBM3 GPUs
Inference hardware	NVIDIA T4, 16 GB VRAM

References

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2. MIC-DKFZ: AutoPET-III LesionTracer and AutoPET-IV LesionLocator, Apache-2.0 software releases (2024–2025)